ThaiSLM Reverse — Result Report v1 (text / speech → Thai Sign Language avatar)

Scope (reverse_text2sign/prompt/prompt_1.md): the reverse of the forward ThaiSLM — a person types or says a sentence (Thai or English) and a 3D avatar signs it in Thai Sign Language (TSL), half-body, on a web page at **port 8081** (`docker compose up --build` at the repository root also brings up the forward part on 8080). Data from the existing S3 lake mirror (`cloud_s3/`), noise removed clip by clip; no sign is ever invented; production-grade (fast, small, tested, containerised); total cloud cost ≤ $5–6. Everything in this report was measured on the deployed stack or with the code in `reverse_text2sign/`.

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0. TL;DR

question	answer
Does it work end to end?	**Yes.** Thai or English text, or speech (Thai / English microphone) → TSL structure → a VRM avatar signs it in the browser at 8081, half-body, with the current sign captioned, a timeline, speed control, an editable gloss list and `.vrma` export for Unity.
How is a sign produced?	**Only from recordings.** 9,694 dictionary signs (TTRS, th-sl.com, TSL51, eye-checked YouTube lessons) and 40 recorded everyday phrases were retargeted from real signers' MediaPipe landmarks onto the avatar. A word without a recorded sign is fingerspelled with recorded letters or shown as text — never generated (test-enforced, §4.5).
Thai ≠ TSL?	**Handled by an explicit TSL structure stage** (time first · object/topic before subject · head-final verb chain · WH last · negation after the predicate · possessor first · Thai-only words dropped · pronoun forms → pointing signs). TSL51 deaf-expert gloss order: **62/62** (development set — see §4.2 caveat); Thai word order would get 14/62.
Everyday sentences covered?	**Thai: 97.6 % of words** become real recorded signs (93.8 % of sentences fully signed, 0 text cards). **English: 97.8 %** (94.3 %), via local NLLB translation. Your example *"วันนี้คุณไปไหน?"* → **วันนี้ · คุณ · ไป · ที่ไหน** (WH-question face).
Is the motion faithful?	**Close to the recordings.** Avatar wrist error median **0.067 shoulder widths**, elbow 0.094, fingertips 0.23 palm lengths. The forward ThaiSLM recogniser (trained on real signers, source clip removed from its bank) names the **avatar's** sign top-1 **52.5 %** vs **63.0 %** for the real signer's own landmarks (n = 1,000) → the avatar keeps **83 %** of the recognisable signal.
Continuous signing?	Clips are cut to their strokes and joined by minimum-jerk transitions; the **tempo is grounded in 28 everyday phrases recorded by a real signer**: our first, citation-speed sentences were 2× too slow (median 4.1 s vs 2.0 s); final 2.6 s. No body-bone pops (≤ 35°/frame, test-enforced), hand contact kept (wai / ขอบคุณ knuckles 1 cm).
Non-manual features?	Each sign carries the recorded signer's own face (brows, mouth, blinks → VRoid morphs) and head. Sentence level: WH-questions raise the brows — **measured** on TSL51 (15 WH vs 237 statements, p = 0.001); yes/no questions use the same marker as a flagged **rule** (no data).
Fast and small?	Thai text → full motion **59 ms p50 / 102 ms p95** at the API (28 kB gzip per sentence); English **0.50 s / 0.74 s**; Thai speech 1.9 s, English speech 0.7 s. Browser: **0.67 ms JavaScript per frame**, avatar 3.5 MB over the wire (20 MB delivered file), app 195 kB.
Speech input accuracy?	English (base.en): **WER 0.5 %**. Thai (Whisper-small-th): **CER 16 %** on synthetic speech — below target; Whisper-medium-th gets 13 % but is 3× slower on CPU (§7.3).
Not done / not proven	**Comprehension by Deaf viewers is not measured** — that is the human study (review pack §8). Palm-to-palm contact still crosses slightly (ขอบคุณ); 6 % of everyday sentences (9 of 149) still contain a fingerspelled word; the TSL grammar rules were developed on the only gold data (TSL51), so their accuracy outside it is unknown until expert review.
Cost	**$0 cloud.** No GPU job, no GCS object — everything ran on the laptop (CPU; the RTX 2050 only for one encoder comparison). Agent (model) usage is billed separately by Anthropic and is not visible to me.

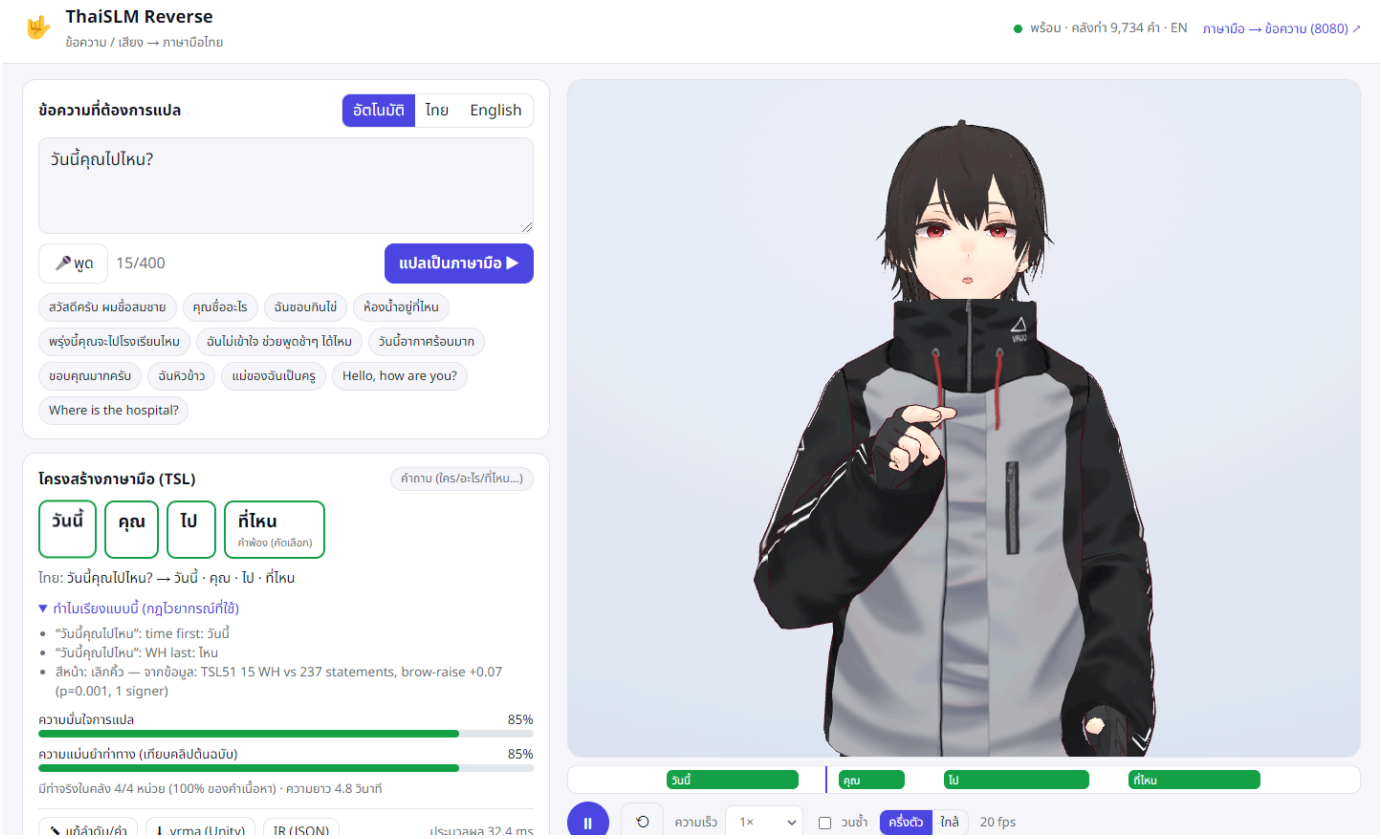


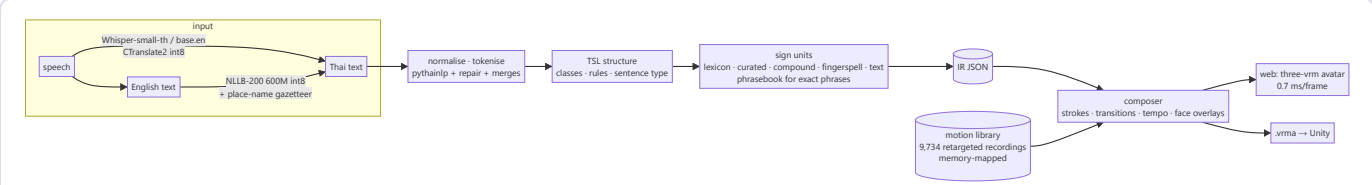
Figure 1 — the production page (8081): input + TSL structure (left), the avatar signing วันนี้ with the current sign captioned (right).

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1. The 8 considerations of the prompt → what the system does

#	consideration	design answer	where
1	Thai ≠ TSL	a separate TSL-structure stage with explicit, logged rules; the IR shows every move (*"WH last: ไหน"*)	`t2s/text/grammar.py`, IR `clauses[].moves`
2	dataset / coverage; standard vs context signs; no invented signs	9.7 k recorded signs + 40 phrases; senses / variants kept (สวัสดี elder vs friend); fallback chain ends in fingerspelling / text	`t2s/data/`, `t2s/text/translate.py`
3	an Intermediate Representation	Thai → TSL structure → sign units (clip, hand, location, movement, face) → motion timeline; editable and re-renderable (`POST /api/render`)	`t2s/ir.py`
4	not only hands	per-sign recorded face (brows / eyes / mouth) + head; sentence brow marking	`retarget.py`, `nmm.py`
5	motion must be correct, smooth transitions	retargeting with body anchors + IK + hinge fingers; contact constraint; de-spike; velocity clamp; data-timed transitions	`t2s/motion/`
6	continuous signing	stroke cutting, hold reduction, minimum-jerk transitions, clause pauses, conversational tempo from real phrases	`compose.py`, `timing.json`
7	unknown words / uncertainty	lemma → curated variant → compound → fingerspelling → text; **two separate confidences** (translation, motion)	IR `units[].confidence`
8	evaluate more than looks	translation (TSL51, everyday coverage), motion (retarget error, recogniser back-translation, renders vs video), system (latency, payload, fps); comprehension = expert pack	§4–§8

2. Blueprint



Why this architecture (and not a generative text-to-motion model). The prompt forbids invented signs and asks for production quality. A generative model (text → poses, as in Progressive Transformers / SignGen-style VQ-VAE + LLM systems) needs large aligned corpora and still produces blurred or hallucinated hand shapes; for Thai there is no such corpus (TSL51 has 62 sentences). The concatenative approach (unit selection → transition smoothing → avatar), also used by recent Libras / BSL production systems, shows only signs a person actually signed, makes every choice inspectable, runs in milliseconds on a CPU and costs nothing to train. Its weak points — co-articulation and tempo — were measured and calibrated on real data (§5.4).

Prior bias (prompt: "must not happen"). No model trained on another sign language enters the motion path (the motion is copied from Thai signers). Exemplar choice has no source preference: with ≥ 3 signers the DTW **medoid** (the way most signers sign it) wins; one vote per signer (TSL51's ~10 takes of one person count once). The only pretrained models are for language (Whisper, NLLB) and their outputs are shown in the IR. An embedding "nearest word" fallback was **evaluated and rejected**: on the lake's 3,187 labelled synonym pairs the

best encoder (bge-m3, AUC 0.80) recalls only 9 % of true synonyms at 90 % precision — it would mostly show a wrong sign.

Unity. Unity is not installed on this machine and installing it needs an account sign-in, so production uses three.js + three-vrm (no plug-in, 195 kB of code, 0.7 ms/frame). Every sentence can be downloaded as `.vrma` **(VRM Animation)** — UniVRM (Unity) and three-vrm play it on any VRM avatar for frame-by-frame inspection.

KPIs defined before building, and where they ended

level	KPI	target	result
translation	TSL51 gloss order (dev)	≥ 90 %	**100 %** (62/62)
translation	everyday words → recorded signs (TH / EN)	≥ 95 %	**97.6 % / 97.8 %**
translation	everyday sentences fully signed (TH / EN)	≥ 90 %	**93.8 % / 94.3 %**
translation	invented signs	0	**0** (test)
motion	recogniser top-1, avatar ÷ real signer	≥ 0.75	**0.83**
motion	wrist retarget error (median)	≤ 0.10 sw	**0.067**
motion	body-bone rotation per frame	< 35°	**met** (test)
motion	sentence tempo vs real signer (median)	within 1.5×	**1.29×** (2.6 s vs 2.0 s)
system	Thai text → motion p50	≤ 100 ms	**59 ms**
system	English text → motion p50	≤ 1 s	**0.50 s**
system	Thai speech CER	≤ 10 %	**16 %** ✗ (§7.3)
system	browser cost per frame	≤ 4 ms	**0.67 ms**
comprehension	Deaf viewers understand	(human study)	**not measured — review pack §8**
cost	cloud	≤ $5–6	**$0**

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3. Phase 1 — data audit (what is in the lake, what is noise)

3.1 Sources and what the reverse direction can use

source	clips	signers	what it is	usable for avatar motion?
TTRS dictionary	5,043	9	studio, blue screen, small signer, hands rest clasped at the belly, 854×480	**yes** — official dictionary
th-sl.com (NADT)	7,719	9	studio, grey, tight 600×600, arms hang at rest	**yes** — official dictionary; incl. the full Thai alphabet (consonants, vowels, tone marks)
TSL51 researcher	547 signs + 252 sentences	1	home recordings 640×480, 51 words, 62 Thai sentences with deaf-expert gloss order	signs: yes (consensus); sentences: evaluation
TSL-ONE-S	4,148	29	landmarks only, 2D, no face	no (no depth / face)
YouTube (v7 harvest)	561	25 channels	lessons, mixed quality	**only eye-checked fill-ins** (§3.4)
YouTube (harvest 2026-10-10)	122	—	targeted lessons + conversational phrases	eye-checked words + recorded phrases (§3.5)

The forward project had already extracted every video with MediaPipe Holistic (`data_prep/pose_mp_raw`: pose 33 x/y/z, hands 21×3, face, 52 blendshapes) — the reverse part reuses these, so **no re-extraction and no GPU** were needed.

3.2 Quality measured per clip, then looked at

`t2s/data/quality.py` measures, on the cleaned landmarks the avatar would actually play: body framing, dominant-hand tracking where the hand is up, longest lost-hand gap, hand swaps between arms, wrist / finger **tracking noise** (median residual after a 0.28 s Savitzky–Golay fit — real fast signing is smooth at that scale, detector flicker is not), hands leaving the picture, zoom, face visibility, sign span. Several first-draft metrics were wrong and were corrected **by looking at the clips**:

- "active signing" by hand height failed on TTRS (hands rest at the belly) → distance from each clip's own rest position;
- "finger jitter" as frame-to-frame acceleration measured real finger motion → residual after smoothing, run-wise, median;
- "hand swap" flagged palm-to-palm signs (wai) → a duplicate hand must also have the **same chirality** (mirror-image hands are real contact);
- face visibility and swaps hard-rejected genuine signs made on the face (หน้ากาก) or with crossing hands (ป้องกัน, มารยาท) → soft rules (tier B, then checked in renders).

Thresholds sit at the ~99th percentile of the clean studio dictionaries; contact sheets per failure reason were reviewed by eye (e.g. th-sl "hand_rate" rejects = motion blur, hands out of frame, a hand on the foot; TTRS "no_signing" = tiny belly-level signs the activity rule misses — 15 clips, left out).

source	tier A	tier B	rejected	top reasons
TTRS	4,501	378	164	lost-hand gap, low hand tracking, swaps
th-sl.com	7,153	356	210	low hand tracking, swaps, hands out of frame
TSL51 signs	490	30	27	hand tracking
YouTube v7	462	37	62	hand tracking, gaps

Cleaning (`t2s/motion/clean.py`) repairs before measuring: duplicate hands dropped, hands re-attached to the right arm, single-frame finger flicker removed, gaps ≤ 0.32 s interpolated — nothing longer is invented.

3.3 Exemplar selection — no source prior

9,694 signs: th-sl 5,507 · TTRS 4,077 · TSL51 23 · YouTube 87. Left-dominant signers (743) are mirrored so every sign is right-dominant — mixing hands inside a sentence looks wrong. 281 concepts had ≥ 3 signers and were chosen by consensus.

3.4 YouTube — "choose carefully"

All 69 YouTube-only candidates were looked at one by one: **43 accepted** (adult signer, clean framing), **26 rejected** — child / scout learners (17), picture overlays or captions over the signing (9). Decisions with reasons: `t2s/data/youtube_fill.csv`.

3.5 Collecting missing words (user-approved harvest)

The gap list came from data (frequent TNC words + the daily vocabulary + the everyday test sentences). TTRS was queried first: 5,045 entries online vs 5,043 held — nothing new. **th-sl.com was not scraped** (its robots.txt reserves rights against AI agents). YouTube: 209 candidates → 122 downloaded (polite, sequential) → same MediaPipe + QC → 109 passed → **eye check: 45 words, 40 recorded conversational phrases, 24 rejected** (children, lesson sentences, overlays, duplicates; `t2s/data/harvest_decisions.csv`). Words that exist only as YouTube compounds never hide the dictionary decomposition (กินข้าว stays ข้าว + กิน, as the TSL51 expert signs it).

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4. Phase 2 — the representation (language side)

4.1 Pipeline

normalise → pythainlp newmm + a local re-split repair (newmm cut บอกว่า as บอ|กว่า) + merges into dictionary entries → closed-class roles first (particles, WH, negation, time, pronouns, modals, copula, classifiers…) → the dictionary's own part of speech → pythainlp tagger as fallback → TSL rules:

rule	evidence
time words first	TSL51 (พรุ่งนี้ …), interpreter practice
object / topic before subject, verb last	TSL51 majority (OSV 36 vs SOV 7) and Thai PBS interpreter interview (*"ข้าว … กิน"*)
verb chain head-final: Thai V1 V2 O → O V2 V1 (ชอบ เที่ยว กรุงเทพ → กรุงเทพ เที่ยว ชอบ)	TSL51
incorporated objects stay with their verb, not fronted (กินข้าว → ข้าว กิน)	TSL51
linking verbs keep their complement (เกิด หูหนวก, ชื่อ แมว)	TSL51
WH last; negation after the predicate	TSL51, sign-language typology
possessor first (แฟนฉัน → ฉัน แฟน)	TSL51
copula / future / politeness words not signed	TSL51
Thai pronoun forms → the pointing sign (ผม/ดิฉัน → ฉัน; ผม = "hair" only before ยาว/สั้น…)	the dictionaries record ผม = hair, มัน = potato, หนู = mouse

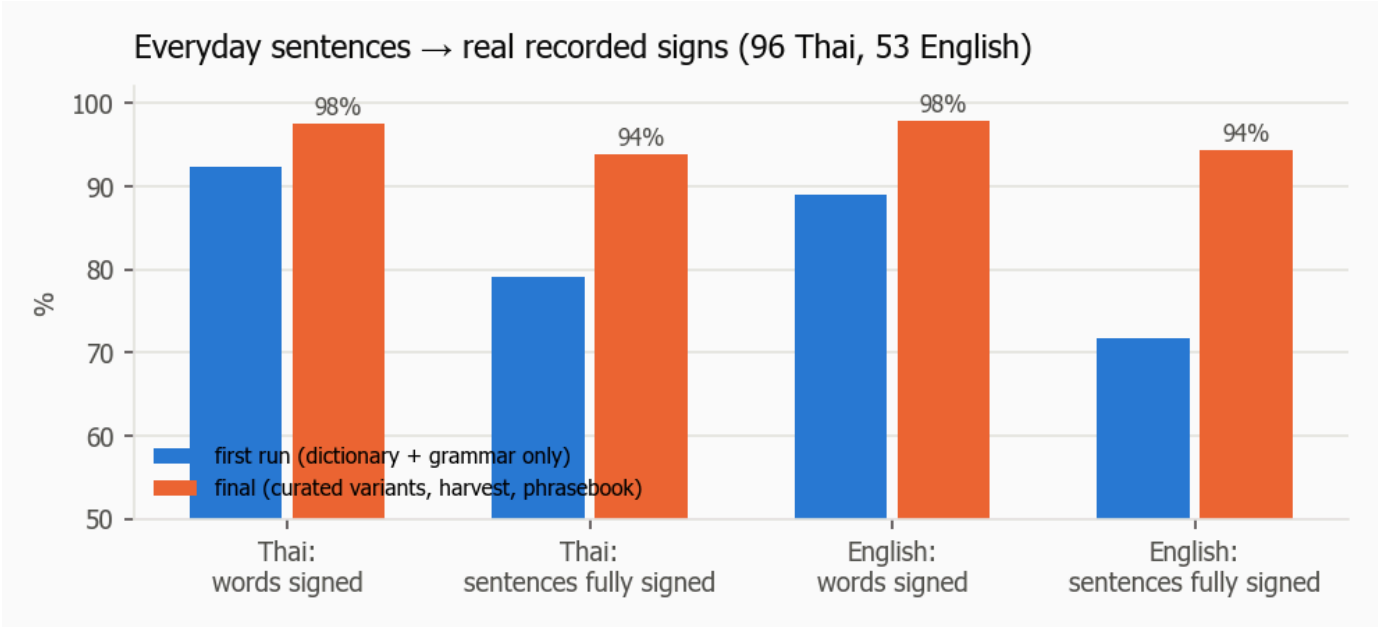
4.2 TSL51 (62 Thai sentences with expert gloss order)

system	exact gloss sequence
Thai word order (word-for-word)	14 / 62
rules without object fronting	26 / 62
final rules	**62 / 62**

Caveat: TSL51 is the only Thai↔TSL parallel data, the rules were written while looking at it, and its 62 sentences come from ~14 templates. It is a development score, not a test: how well the rules generalise must be judged by Deaf experts (§8).

4.3 Everyday conversation (user request: "cover Thai and English, e.g. วันนี้คุณไปไหน?")

`tests/conversation.tsv`: 96 Thai and 53 English everyday sentences (greetings, introductions, where/when/what, food, shopping, help, feelings, family, work, weather, travel). The first run exposed gaps that a dictionary alone cannot close; each was fixed as a general rule, never per sentence: colloquial forms mapped to recorded signs in one reviewable table (`t2s/text/curated.py`: ไหน → ที่ไหน, เจอกัน → เจอ, ไปไหนมา → ไป ที่ไหน, หรือยัง → ยัง + question, กี่โมง → เวลา เท่าไร …, each marked `curated`, confidence 0.85), degree adverbs after the predicate, possessive pronoun signs split, benefactive ให้ / expletive มัน / ใน after location verbs dropped, a recorded phrasebook for exact matches, a place-name gazetteer before English MT (NLLB wrote "Chiang Mai" as ชิงแมย์).



	first run	final
Thai: words → recorded signs	92.4 %	**97.6 %**
Thai: sentences fully signed	79.2 %	**93.8 %**
English: words → recorded signs	89.0 %	**97.8 %**
English: sentences fully signed	71.7 %	**94.3 %**

Examples (final): *วันนี้คุณไปไหน* → วันนี้ คุณ ไป ที่ไหน · *ฉันไม่ชอบกินผัก* → ผัก ฉัน กิน ไม่ชอบ · *ช่วยพูดช้าๆหน่อยได้ไหม* → ช้า พูด ช่วย ได้ (yes/no) · *Hello, how are you?* → สวัสดี คุณ สบายดี (yes/no) · *I want to travel to Chiang Mai.* → เชียงใหม่ ฉัน ไป อยาก. Remaining fingerspelled words (9 of 389 units): names (สมชาย, จอห์น), ทำ (its only dictionary clip is a truncated file, in S3 too), เสื้อ (only a child learner's clip exists — rejected), ราคา / ลดราคา, โมง in clock times.

4.4 The IR (abridged, for *ผมชื่อจอห์น*)


```
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   "manual": {"dominant": "right", "two_handed": true, "location": "chest", "movement": 1.11, "movement_kind": "path"},
   "confidence": {"translation": 1.0, "motion": 0.852}},
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   "note": "ไม่มีท่าในคลัง — สะกดนิ้ว (no recorded sign — fingerspelled); ไม่มีท่าของเครื่องหมาย อ์ — ข้าม",
   "confidence": {"translation": 0.6, "motion": 0.92}}],
 "confidence": {"translation": 0.6, "motion": 0.852, "coverage": 0.667}}
```


(exact output of `python main.py translate "ผมชื่อจอห์น"`, trimmed to these fields; `consensus` is null when fewer than 3 signers exist; the ์ mark of จอห์น has no usable recorded clip and is skipped, as the note says)

4.5 No invented signs (test-enforced)

`tests/test_language.py::test_no_invented_signs` : for unseen inputs (a name, a long noun phrase, an out-of-domain technical sentence, a letter string) every `sign` unit is a library recording, every `fingerspell` letter is a recorded letter, everything else is a `text` card. The translator has no code path that can emit a motion it was not given.

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5. Phase 3–4 — grounding the motion in the data, and the avatar

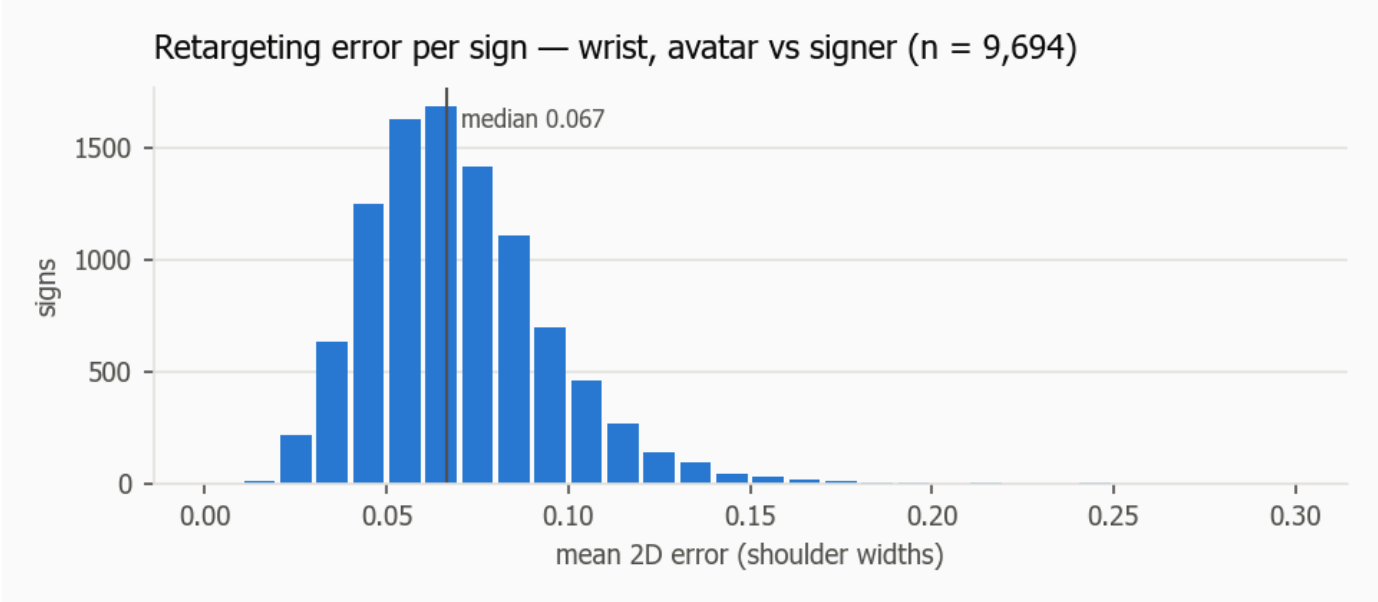
5.1 Retargeting (MediaPipe → VRM, offline, CPU, ≈ 0.1–0.4 s per clip)

- model frame = the VRM 0.x rig (+Y up, faces −Z) so the browser writes quaternions to normalized bones unchanged;
- depth: MediaPipe z rescaled per clip by the factor that makes upper arm / forearm **rigid** (min. coefficient of variation);
- torso / head relative to the clip's own rest frames (camera angle removed), damped;
- wrist target placed relative to the **nearest body anchor** (own shoulder, mouth) and scaled by shoulder width — the VRoid head is large and its shoulders narrow, so copying joint angles would miss the chin / chest; kept in front of the torso surface;
- two-bone IK with the signer's own elbow as pole; forearm twist shared with the wrist;
- palm frame from wrist / middle / index / pinky knuckles (axes smoothed over tracked runs — an edge-on palm otherwise flips);
- fingers: proximal swing without twist, intermediate / distal as hinges, flexion unwrapped (a tight curl past 180° stays a curl);
- two-hand contact: when the signer's wrists are close, both avatar wrists keep the signer's relative vector (wai, clapping);
- face: 52 blendshapes relative to the signer's neutral face → VRoid morphs (brows surprised / angry / sorrow, blinks, wide eyes, mouth A/I/U/O, smile, frown);
- clean-up: quaternion de-spike, then a velocity clamp (30°/frame ≈ 750°/s) on body / arm / wrist bones.

Two defects were found by the tests and fixed in the library before the final build: 26 % of signs had a wrist / forearm jump

> 30° in one frame (palm-frame flips; up to 85°) and fingers flickered between fully bent and straight (hinge angle wrap-around).

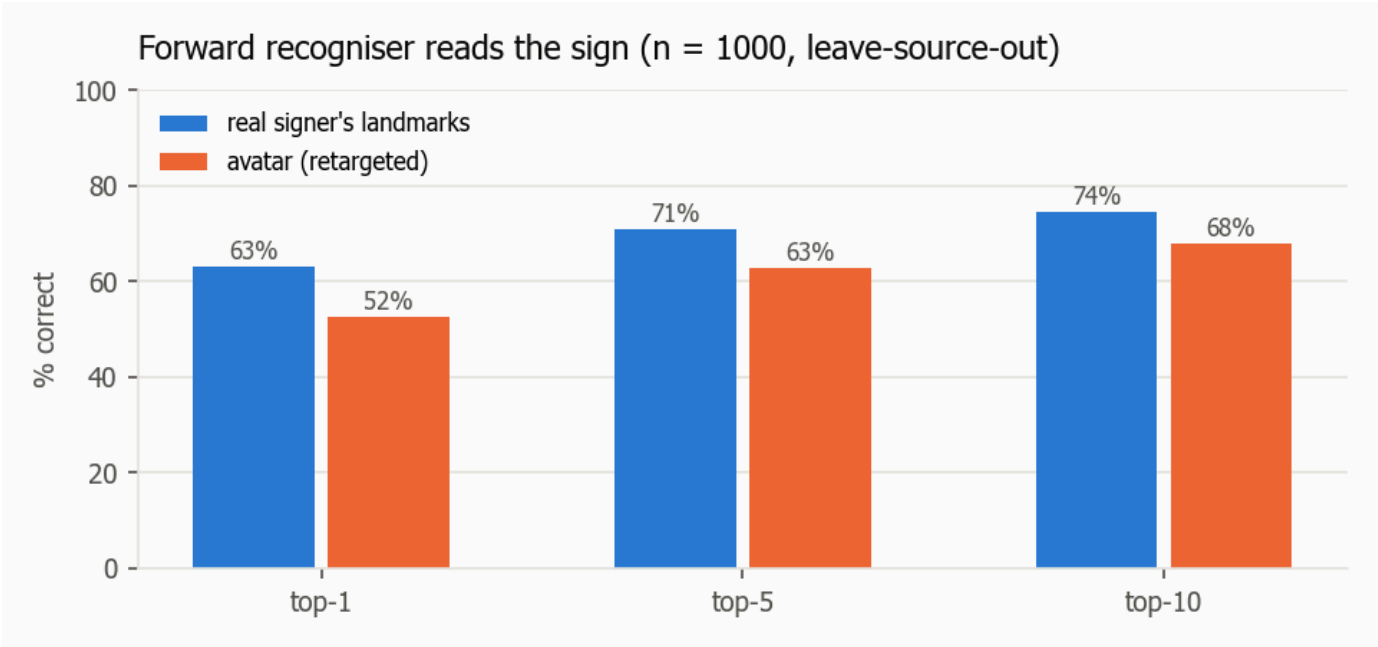
5.2 Fidelity — the avatar against the signer



measure (2D, same normalised frame)	median	p90
wrist (shoulder widths)	**0.067**	0.102
elbow (shoulder widths)	0.094	—
fingertips relative to the wrist (palm lengths)	0.232	0.379

5.3 Does the avatar sign look like the sign? (back-translation)

The forward ThaiSLM v8 recogniser (Thai-only landmark encoder) reads landmarks generated from the avatar's skeleton and names the sign among ~9.8 k concepts; the source clip is removed from its bank, so the avatar must look like **other** signers' versions.



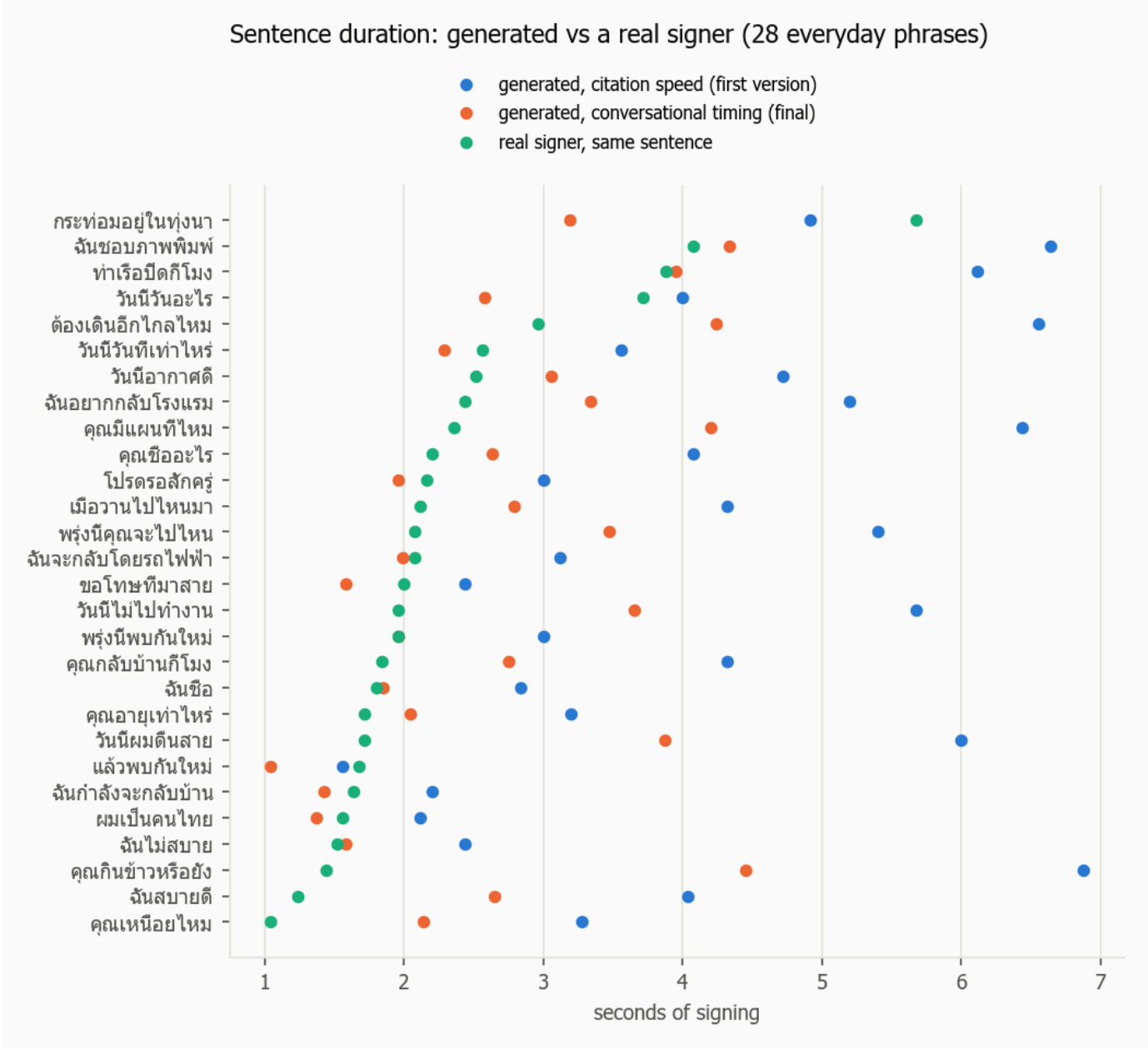
n = 1,000	top-1	top-5	top-10
real signer's landmarks (upper bound)	63.0 %	70.9 %	74.5 %
avatar	**52.5 %**	**62.7 %**	**67.8 %**

The avatar keeps 83 % of the top-1 signal (median cosine avatar ↔ human embedding 0.84). It is a machine proxy: a Deaf viewer may weigh hand shape, face and movement differently.

5.4 Continuous signing and tempo

Clips are cut to their **stroke** (between the first and last moment the hands are at signing distance, snapped to a hold), joined by per-bone slerp with a **minimum-jerk** profile (zero velocity at both holds), face morphs cross-faded; rest → first sign and last sign → rest; a short hold between clauses; held endings inside a sentence capped. Timing evidence:

measurement	result	used for
reach (lift) time of 1,965 dictionary clips	**0.44–0.46 s regardless of distance** (0.3–3 shoulder widths)	first transition model
TSL51 lab sentences vs the same signer's isolated words	sentences **1.55× slower** (deliberate recording)	rejected as a tempo reference
28 everyday phrases recorded by an adult signer vs our generation of the same Thai sentence	real = **0.58 ×** citation strokes + **0.21 ×** our transitions (NNLS fit)	**final**: strokes ×1.5, transitions ≈ 0.25 s, holds ≤ 0.12 s



Median sentence: real signer 2.04 s · first version 4.06 s · **final 2.64 s**. Applied a little gentler than measured — an avatar is read, not overheard; the viewer can change speed (0.5–1.5×).

5.5 Comparison with recordings (the prompt's "compare with the video")



Figure — top of each pair: the dictionary video; bottom: the avatar at the same moments of the sign. ขอบคุณ (wai then opening), ห้องน้ำ (pinch at the nose, other hand at the belly), ชื่อ (two-handed tap), หน้ากาก (hand over

the face, a tier-B sign rescued by the soft face rule).

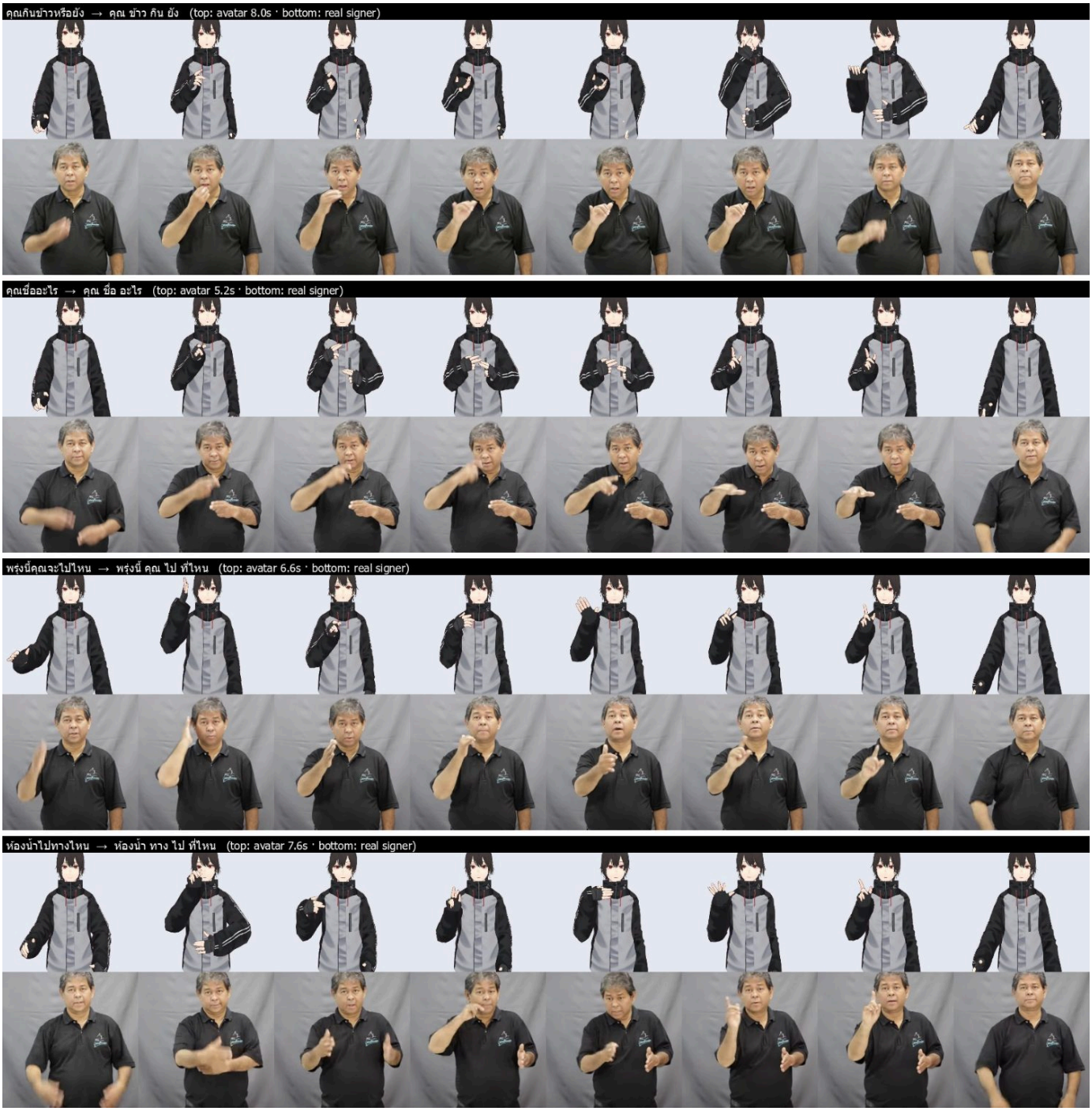


Figure — top: the avatar generating the sentence from words (phrasebook off); bottom: a real signer's recording of the same sentence. Word order matches in these examples; the real signer is faster and sometimes uses another variant (อะไร: open palm vs our consensus index-finger variant).

Visible limits: at palm-to-palm moments (ขอบคุณ) the avatar's hands still cross diagonally instead of pressing flat; the anime avatar's large sleeves hide part of the forearm; mouthing follows the dictionary signer, not the sentence.

5.6 The avatar on the page

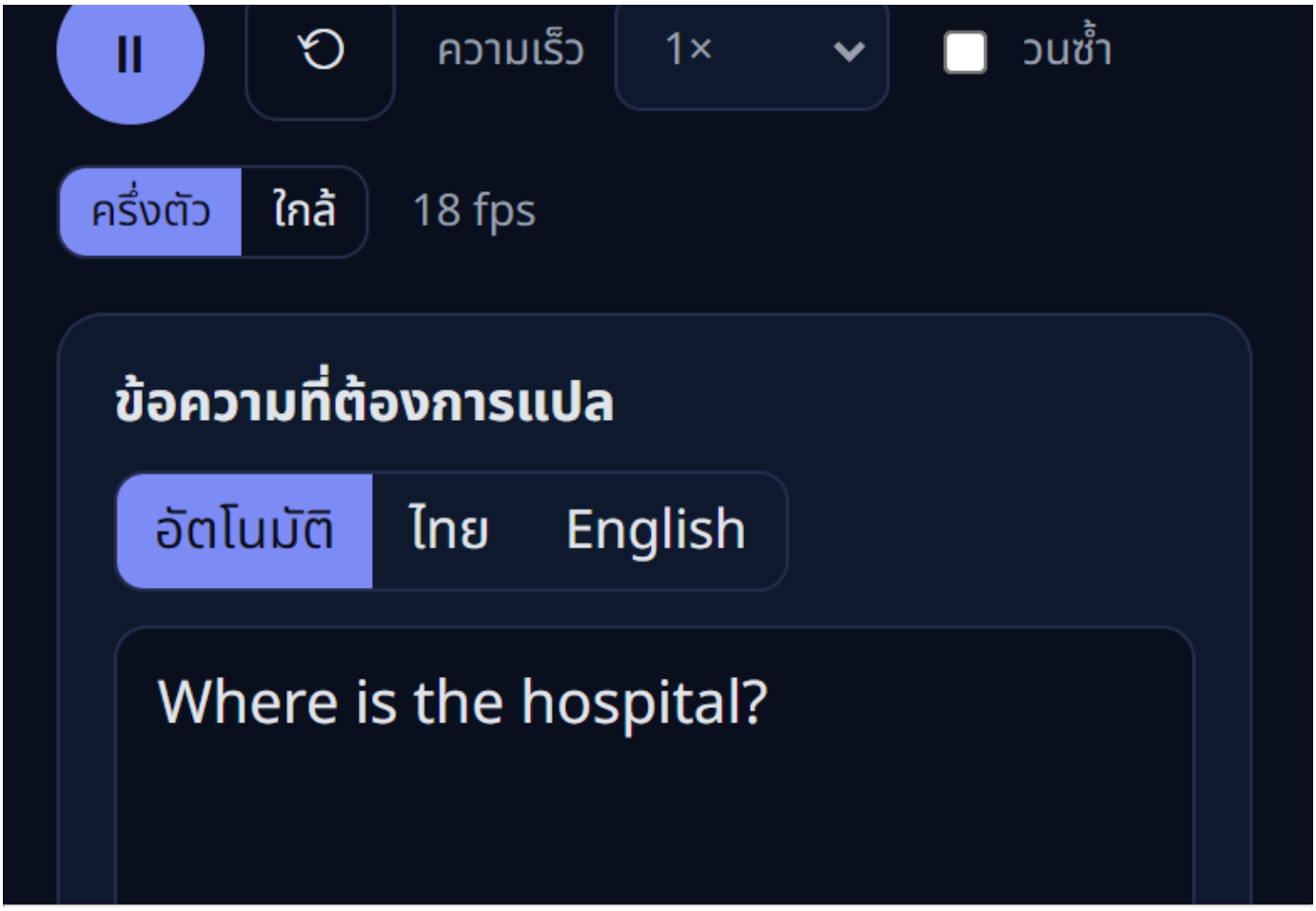
`avatar.vrm` = the delivered VRoid model with textures capped at 2048 px (19.5 → 9.5 MB, 3.5 MB gzip on the wire; same skeleton and morphs). Half-body camera framing that keeps a 1 m wide signing space visible (0.8 m on phones); the avatar fills the right 2/3 of a desktop screen. Breathing and blinking when idle; the timeline is

played at display rate with slerp between the 25 fps frames. Mobile: the page scrolls to the avatar after translating.

โรงพยาบาล



โรงพยาบาล | อยู่ | ที่ไหน



6. Phase 5 — services (`docker compose up --build` → :8081)

service	image	role
`reverse-web`	nginx + built Vite app (49 MB)	the page, `/api` proxy, immutable asset caching, gzip (incl. the VRM)
`reverse-api`	python:3.11-slim-bookworm + numpy / scipy / pythainlp / FastAPI (450 MB)	text → IR → motion; memory-mapped library (`models/` mounted read-only); 2 workers; LRU cache
`reverse-speech`	CTranslate2 int8 models converted at build time from the official checkpoints (1.6 GB)	`/asr` (Thai, English), `/mt` (EN → TH)

API: `/api/translate`, `/api/render` (edited IR), `/api/sign/{word}`, `/api/lookup`, `/api/asr`, `/api/export/vrma`, `/api/rest`, `/api/health`. Wire format: int16 quaternions + uint8 morphs, base64 in JSON, gzip (≈ 28 kB per sentence). Engineering notes: the CTranslate2 4.5 wheel needs `-bookworm` base images (Debian trixie refuses its executable stack); the library is written to a new directory and swapped in (Windows cannot overwrite a memory-mapped file while the API runs).

7. Phase 6 — system measurements

7.1 Latency and size (deployed stack, laptop CPU, quiet machine)

path	p50	p95	response
Thai text → IR + motion (through nginx)	**59 ms**	102 ms	28 kB gzip
English text → NLLB → IR + motion	**497 ms**	742 ms	19 kB
Thai speech (≈ 2 s) → text	1.9 s	2.1 s	—
English speech → text	0.7 s	—	—
browser: JavaScript per frame	**0.67 ms**	—	(GPU excluded)
first load	app 195 kB + avatar 3.5 MB		

Software-only WebGL (headless, no GPU) renders ~10–25 fps; with any GPU the page is limited by the display rate.

7.2 Tests

`python main.py test` — 23 tests (language: TSL51 regression, the conversation examples incl. the user's, pronoun homographs, no invented signs, separate confidences, phrasebook; motion: finite unit quaternions, rest at both ends, no body-bone pops, no finger flicker, ordered timings, speed control, `.vrma` validity; API: health, translate + edited re-render, errors, lookup, sign, VRMA). All pass.

7.3 Speech input (synthetic speech: an upper bound, not field accuracy)

	model	error	exact	latency
English (53 sentences, Windows "Zira")	Whisper base.en	**WER 0.5 %**	52/53	0.7 s
Thai (96 sentences, VachanaTTS)	Whisper-small-th (deployed)	**CER 16 %**	64/96	1.9 s
Thai, same audio, local A/B	Whisper-medium-th	CER 13 %	71/96	6.7 s

Fixes made: very short replies (ใช่, ได้, ฝนตก) were silenced by the voice-activity filter → retried without it (0 empty results now); a more sensitive VAD and shorter decoding windows were tried and rejected (CER 17–26 %). Recommendation: switch to Whisper-medium-th if a GPU or a 6–7 s budget is acceptable (one line in `service/speech/Dockerfile`). Every recognised text is shown and editable before signing.

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8. What a Deaf reviewer must still judge (comprehension level) — the review pack

Machine proxies cannot prove understanding. Prepared for the experts:

- `result_reporting/review/conversation.csv` — the 149 everyday sentences with our TSL order, fallbacks and confidences (translation level);
- the live page — every sentence can be edited (✎) and re-rendered, and exported as `.vrma` for frame-by-frame review in Unity;

- `result_reporting/review/renders/` — sign sheets (video vs avatar) and sentence sheets (generated vs a real signer);
- items to rate first: object-fronting default (TSL51 vs Thai PBS both support it, but variation exists), the yes/no brow raise (a rule without data), the 39 curated variants and 23 fixed expressions in `t2s/text/curated.py`, the consensus choice for multi-variant signs (อะไร, สวัสดี).

Suggested protocol: 3 levels per sentence — translation (is this the TSL you would sign?), motion (hand shape / place / movement correct?), comprehension (watch once without text, write what was signed).

9. Cost

item	used	cost
GCP Vertex AI / Compute	none (no job submitted)	$0
GCS	no object written	$0
AWS S3	read only (one `ls` of the lake)	≈ $0
local	CPU (retargeting 9.7 k clips ≈ 8 min × 3 builds, stroke annotation ≈ 8 min); GPU only for the encoder comparison	electricity

GPU training was never needed: the motion comes from recordings, the language stage is rules + pretrained translators, and the heaviest offline step (retargeting) is CPU-parallel. A Spot A100 would not have shortened anything that took longer than minutes.

10. Limits and next steps (honest)

1. **Comprehension unmeasured** — run the §8 study with Deaf signers before any public use.
2. **Grammar generalisation unknown** beyond TSL51 — expert review of `review/conversation.csv`; then encode their corrections as rules.
3. **Palm-to-palm contact** still crosses (ขอบคุณ); next: orientation constraint between the two palms during contact frames.
4. **Thai ASR CER 16 %** — Whisper-medium-th (13 %) or a GPU; real microphone tests.
5. **Coverage gaps** — ทำ, เสื้อ, ราคา, ลดราคา, จริง, ทุก: no usable adult recording exists in any source; a short recording session with a Deaf signer (≈ 50 everyday words) would close most remaining fingerspelling.
6. **Mouthing / non-manual grammar** — negation head-shake and topic marking are not modelled beyond each clip's own face.
7. **Licensing** — th-sl.com and YouTube material: research use only; a public release needs the rights holders' consent.

11. Reproduce

```
cd reverse_text2sign
python main.py audit && python main.py select && python main.py library && python main.py lexicon && python main.py
timing
python main.py avatar && python main.py test && python main.py eval all
cd .. && docker compose up --build      # → http://localhost:8081
```

File map: `reverse_text2sign/README.md`. Data: `cloud_s3/` (S3 lake mirror; the reverse harvest is in `cloud_s3/agent_data/reverse_harvest/`); derived data in `reverse_text2sign/data/` (git-ignored, rebuilt by the commands above). Expert review pack: `result_reporting/review/`.